A QUASI-REGULAR SEQUENCE FOLLOWED BY A SEQUENCE THAT IS H1-REGULAR MODULO THE IDEAL IT GENERATES IS QUASI-REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of the cards already published from this chapter, so that the batch selector treats them as one source instead of deferring them as a same-source clash.

Allowed standard prerequisites (editorial).

- Quasi-regularity of a sequence and H_1-regularity of a sequence, the two regularity notions of the section. defined in the discussion of the same section, source lines 7100-7298, before the counted statement. Supplies the hypothesis on f_1 to f_n, the hypothesis on the images of g_1 to g_m, and the goal for the juxtaposed sequence.
- For a quasi-regular sequence, consecutive quotients of powers of the ideal are direct sums of copies of the quotient by the ideal. used at proof lines 7540-7542 where I^{d-1}/I^d is identified as a direct sum of copies of A/I. Makes the first complex in the displayed short exact sequence a direct sum of Koszul complexes over A/I, hence acyclic in degree one.
- The intersection lemma relating an ideal, a sequence and its powers. the author cites this at proof lines 7566-7569 as Lemma lemma-H1-regular-in-quotient (tag 0665), the preceding key lemma of the same section. Converts the membership obtained after applying quasi-regularity of the g's into the membership in I^{d-e}(g_1, ..., g_m) needed for the final step.

Source reference. Original lemma 0667: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7926–7926. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 062I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7797–7797. Navigation pointer to the chapter's lemma identifying H_1-regularity with quasi-regularity, which the author uses to pass between the two notions; the inductions over the powers I^d remain inside the counted proof span.

Source reference. Original lemma 0665: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7924–7924. Navigation pointer to the immediately preceding intersection lemma of the same chapter, invoked at the last step of the proof; that statement is counted by the sibling card S1-algebra-H1-regular-in-quotient of this lane and nothing here re-proves it.

Lemma 0.2. *Let A be a ring. Let I be an ideal generated by a quasi-regular sequence $f_1, \ldots, f_n$ in A. Let $g_1, \ldots, g_m \in A$ be elements whose images $\overline{g}_1, \ldots, \overline{g}_m$ form an H_1-regular sequence in A/I. Then $f_1, \ldots, f_n, g_1, \ldots, g_m$ is a quasi-regular sequence in A.*

Proof. We claim that $g_1, \ldots, g_m$ forms an H_1-regular sequence in A/I^d for every d. By induction assume that this holds in A/I^{d-1}. We have a short exact sequence of complexes

$$0 \to K_\bullet(A, g_\bullet) \otimes_A I^{d-1}/I^d \to K_\bullet(A/I^d, g_\bullet) \to K_\bullet(A/I^{d-1}, g_\bullet) \to 0$$

Since $f_1, \ldots, f_n$ is quasi-regular we see that the first complex is a direct sum of copies of $K_\bullet(A/I, g_1, \ldots, g_m)$ hence acyclic in degree 1. By induction hypothesis the last complex is acyclic in degree 1. Hence also the middle complex is. In particular, the sequence $g_1, \ldots, g_m$ forms a quasi-regular sequence in A/I^d for every $d \geq 1$, see Lemma 062I. Now we are ready to prove that $f_1, \ldots, f_n, g_1, \ldots, g_m$ is a quasi-regular sequence in A. Namely, set $J = (f_1, \ldots, f_n, g_1, \ldots, g_m)$ and suppose that (with multinomial notation)

$$\sum\nolimits_{|N|+|M|=d} a_{N,M} f^N g^M \in J^{d+1}$$

for some $a_{N,M} \in A$. We have to show that $a_{N,M} \in J$ for all N, M. Let $e \in \{0, 1, \ldots, d\}$. Then

$$\sum\nolimits_{|N|=d-e,\ |M|=e} a_{N,M} f^N g^M \in (g_1, \ldots, g_m)^{e+1} + I^{d-e+1}$$

Because $g_1, \ldots, g_m$ is a quasi-regular sequence in A/I^{d-e+1} we deduce

$$\sum\nolimits_{|N|=d-e} a_{N,M} f^N \in (g_1, \ldots, g_m) + I^{d-e+1}$$

for each M with $|M| = e$. By Lemma 0665 applied to I^{d-e}/I^{d-e+1} in the ring A/I^{d-e+1} this implies $\sum_{|N|=d-e} a_{N,M} f^N \in I^{d-e}(g_1, \ldots, g_m)$. Since $f_1, \ldots, f_n$ is quasi-regular in A this implies that $a_{N,M} \in J$ for each N, M with $|N| = d - e$ and $|M| = e$. This proves the lemma. □